

ASCII MASTER FLOW – PANEL 1/12: Powertrain source-of-electricity map

BEV -> battery stores all traction electricity

HEV -> engine + regenerative braking charge non-plug-in battery

PHEV -> external plug + engine backup

FCEV -> hydrogen fuel cell generates onboard electricity

TEST -> where does traction electricity originate?

ASCII MASTER FLOW – PANEL 2/12: Cell to pack systems hierarchy

CELL -> electrochemical unit

MODULE -> grouped cells + mechanical integration

PACK -> modules/cells + enclosure + busbars + sensing

CONTROL -> BMS + contactors + protection

THERMAL -> cooling/heating + propagation barriers

ASCII MASTER FLOW – PANEL 3/12: Lithium-ion discharge rail

ANODE -> releases Li^+ during discharge

ELECTROLYTE -> carries ions

SEPARATOR -> blocks direct electronic contact

CATHODE -> receives Li^+ during discharge

EXTERNAL CIRCUIT -> electrons power motor

ASCII MASTER FLOW – PANEL 4/12: Battery chemistry decision matrix

NMC / NCA -> nickel-cobalt exposure; higher-density designs

LFP -> no nickel or cobalt; different density/safety trade-off

SODIUM-ION -> changes lithium dependence

SOLID-STATE -> solid electrolyte; pre-commercial boundary

RULE -> chemistry decides mineral bill and engineering trade-off

ASCII MASTER FLOW – PANEL 5/12: Energy and power axes

ENERGY DENSITY -> how much energy per mass/volume

POWER DENSITY -> how quickly energy is delivered/accepted

RANGE / WEIGHT -> energy axis

ACCELERATION / REGEN / FAST CHARGE -> power axis

TRAP -> one metric cannot certify whole-pack performance

ASCII MASTER FLOW – PANEL 6/12: AC and DC charging conversion map

GRID AC -> vehicle onboard charger -> regulated battery DC

GRID AC -> offboard fast charger -> regulated battery DC

LEVEL -> power and use context

CONNECTOR -> physical/electrical interface

PROTOCOL -> communication and control

ASCII MASTER FLOW – PANEL 7/12: Charging governance institution map

MINISTRY OF POWER -> charging guidelines

BIS -> connector and safety standards

CEA -> electricity-system and electrical safety regulation

MORTH / CMVR -> vehicle approval and safety

BEE -> efficiency role

ASCII MASTER FLOW – PANEL 8/12: BMS and thermal-control loop

SENSE -> voltage | current | temperature

ESTIMATE -> operating state and imbalance

ACT -> balance | limit | isolate

COOL / HEAT -> keep cells in suitable band

FEEDBACK -> safer operation and slower degradation, not immunity

ASCII MASTER FLOW – PANEL 9/12: Degradation and safety fault tree

HEAT / HIGH SOC / DEEP CYCLING / HIGH RATE -> ageing pressure

DEFECT / ABUSE / SHORT -> local heat generation

SELF-HEATING > HEAT REMOVAL -> thermal runaway

CELL EVENT -> possible module/pack propagation

RESPONSE -> prevent + detect + contain + emergency action

ASCII MASTER FLOW – PANEL 10/12: Hydrogen mobility chain

PRODUCTION ROUTE -> green / blue / grey label

COMPRESSION / TRANSPORT / STORAGE -> upstream burden

PEM FUEL CELL -> electrochemical electricity

ELECTRIC MOTOR -> traction

DIRECT EXHAUST -> water + heat; lifecycle remains wider

MUST REMEMBER: Electric mobility separates BEV, HEV, PHEV and fuel-cell architectures;...

ASCII MASTER FLOW – PANEL 11/12: Alternative-fuel portfolio

ETHANOL / FLEX-FUEL -> petrol-blend engine pathway

BIODIESEL -> diesel-blend pathway

CBG / SATAT -> purified biogas pathway

CNG / LNG / HCNG -> gaseous combustion fuels

HYDROGEN FUEL CELL -> electrochemical electric-drive pathway

CLOSE DISTINCTION: Cell is not pack, kW is not kWh, battery capacity is not vehicle...

ASCII MASTER FLOW – PANEL 12/12: India transition and evidence ladder

TAILPIPE -> GRID -> MANUFACTURE -> END OF LIFE

MINERALS -> CHEMISTRY -> RECYCLING / EPR

PM E-DRIVE -> demand/ecosystem instrument

PLI-AUTO / PLI-ACC / PM-eBUS SEWA / NGHM PILOTS -> distinct routes

APPROVAL -> NOTIFICATION -> AWARD -> COMMISSIONING -> DEPLOYMENT -> OUTCOME

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State vehicle/fuel pathway, chemistry,...